

Zadání 1. úkolu do předmětu IZU

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Pomocí metody UCS najdete nejkratší cestu v mapě složené z pravidelných buněk, kde cena přechodu mezi dvěma stavy (buněkami) je dána číslem, uvedeným v Tabulce 1 (a je stejná pro všechny přechody ze sousedních míst do příslušné buňky). Nepřekročitelné buňky mají hodnotu „Z“ (jako „zed“). Po každém kroku vypíšte nové hodnoty seznamů Open a Closed. Uzly generujte v pořadí zleva doprava a shora dolů, uvažujte 8-okolí buňky (tzn. operátory $\nwarrow, \uparrow, \nearrow, \leftarrow, \rightarrow, \swarrow, \downarrow, \searrow$). Výslednou cestu zapište do tabulky Výsledná cesta. Uzel se skládá ze souřadnic, z ohodnocení a ze souřadnic uzlu, ze kterého byl vygenerován (aby bylo možné nalézt cestu od startu k cíli).

Uzly zapisujte: ([sloupec, řádek], ohodnocení, [souřadnice otcovského uzlu])

Start: ([6, 6], 0, [null])
Cíl: ([3, 3], X, [?, ?])

Výsledná cesta:

[6, 6], [6, 5], [5, 4], [4, 5], [3, 4], [3, 3]

y/x	0	1	2	3	4	5	6	7	8	9
0	9	8	7	7	9	Z	8	8	8	8
1	9	7	8	7	8	Z	5	7	7	7
2	8	7	Z	Z	Z	Z	7	5	6	6
3	9	8	Z	2	9	Z	8	8	9	9
4	7	9	Z	4	9	3	3	3	3	3
5	3	3	3	3	3	Z	4	5	Z	9
6	6	9	9	9	8	Z	2	6	Z	9
7	7	9	Z	Z	Z	Z	Z	Z	Z	9
8	9	9	9	9	9	Z	9	9	9	9
9	9	9	9	9	9	Z	7	6	8	8

Tabulka 1: Mapa přechodů. Např. cena přechodu do cílové buňky je rovna 2 pro všechny buňky s cílovou buňkou sousedící.

0. iterace

Open:

([6, 6], 0, NULL)

Closed:

1. iterace

Open:

([6, 5], 4, [6, 6]), ([7, 5], 5, [6, 6]), ([7, 6], 6, [6, 6])

Closed:

([6, 6], 0, null)

2. iterace

Open:

([7, 5], 5, [6, 6]), ([7, 6], 6, [6, 6]), ([5, 4], 7, [6, 5]), ([6, 4], 7, [6, 5]), ([7, 4], 7, [6, 5])

Closed:

([6, 6], 0, null), ([6, 5], 4, [6, 6])

3. iterace

Open:

[(7, 6], 6, [6, 6]), ([5, 4], 7, [6, 5]), ([6, 4], 7, [6, 5]), ([7, 4], 7, [6, 5]), ([8, 4], 8, [7, 5])

Closed:

[(6, 6], 0, null), ([6, 5], 4, [6, 6]), ([7, 5], 5, [6, 6])

4. iteraçe

Open:

[(5, 4], 7, [6, 5]), ([6, 4], 7, [6, 5]), ([7, 4], 7, [6, 5]), ([8, 4], 8, [7, 5])

Closed:

[(6, 6], 0, null), ([6, 5], 4, [6, 6]), ([7, 5], 5, [6, 6]), ([7, 6], 6, [6, 6])

5. iteraçe

Open:

[(6, 4], 7, [6, 5]), ([7, 4], 7, [6, 5]), ([8, 4], 8, [7, 5]), ([4, 3], 16, [5, 4]), ([6, 3], 15, [5, 4]), ([4, 4], 16, [5, 4]), ([4, 5], 10, [5, 4])

Closed:

[(6, 6], 0, null), ([6, 5], 4, [6, 6]), ([7, 5], 5, [6, 6]), ([7, 6], 6, [6, 6]), ([5, 4], 7, [6, 5])

6. iteraçe

Open:

([7, 4], 7, [6, 5]), ([8, 4], 8, [7, 5]), ([4, 3], 16, [5, 4]), ([6, 3], 15, [5, 4]), ([4, 4], 16, [5, 4]), ([4, 5], 10, [5, 4]), ([7, 3], 15, [6, 4])

Closed:

([6, 6], 0, null), ([6, 5], 4, [6, 6]), ([7, 5], 5, [6, 6]), ([7, 6], 6, [6, 6]), ([5, 4], 7, [6, 5]), ([6, 4], 7, [6, 5])

7. iterace

Open:

([8, 4], 8, [7, 5]), ([4, 3], 16, [5, 4]), ([6, 3], 15, [5, 4]), ([4, 4], 16, [5, 4]), ([4, 5], 10, [5, 4]), ([7, 3], 15, [6, 4]), ([8, 3], 16, [7, 4])

Closed:

([6, 6], 0, null), ([6, 5], 4, [6, 6]), ([7, 5], 5, [6, 6]), ([7, 6], 6, [6, 6]), ([5, 4], 7, [6, 5]), ([6, 4], 7, [6, 5]), ([7, 4], 7, [6, 5])

8. iterace

Open:

([4, 3], 16, [5, 4]), ([6, 3], 15, [5, 4]), ([4, 4], 16, [5, 4]), ([4, 5], 10, [5, 4]), ([7, 3], 15, [6, 4]), ([8, 3], 16, [7, 4]), ([9, 3], 17, [8, 4]), ([9, 4], 11, [8, 4]), ([9, 5], 17, [8, 4])

Closed:

([6, 6], 0, null), ([6, 5], 4, [6, 6]), ([7, 5], 5, [6, 6]), ([7, 6], 6, [6, 6]), ([5, 4], 7, [6, 5]), ([6, 4], 7, [6, 5]), ([7, 4], 7, [6, 5]), ([8, 4], 8, [7, 5])

9. iterace

Open:

([4, 3], 16, [5, 4]), ([6, 3], 15, [5, 4]), ([4, 4], 16, [5, 4]), ([7, 3], 15, [6, 4]), ([8, 3], 16, [7, 4]), ([9, 3], 17, [8, 4]), ([9, 4], 11, [8, 4]), ([9, 5], 17, [8, 4]), ([3, 4], 14, [4, 5]), ([3, 5], 13, [4, 5]), ([3, 6], 19, [4, 5]), ([4, 6], 18, [4, 5])

Closed:

([6, 6], 0, null), ([6, 5], 4, [6, 6]), ([7, 5], 5, [6, 6]), ([7, 6], 6, [6, 6]), ([5, 4], 7, [6, 5]), ([6, 4], 7, [6, 5]), ([7, 4], 7, [6, 5]), ([8, 4], 8, [7, 5]), ([4, 5], 10, [5, 4])

10. iterace

Open:

([4, 3], 16, [5, 4]), ([6, 3], 15, [5, 4]), ([4, 4], 16, [5, 4]), ([7, 3], 15, [6, 4]), ([8, 3], 16, [7, 4]), ([9, 3], 17, [8, 4]), ([9, 5], 17, [8, 4]), ([3, 4], 14, [4, 5]), ([3, 5], 13, [4, 5]), ([3, 6], 19, [4, 5]), ([4, 6], 18, [4, 5])

Closed:

([6, 6], 0, null), ([6, 5], 4, [6, 6]), ([7, 5], 5, [6, 6]), ([7, 6], 6, [6, 6]), ([5, 4], 7, [6, 5]), ([6, 4], 7, [6, 5]), ([7, 4], 7, [6, 5]), ([8, 4], 8, [7, 5]), ([4, 5], 10, [5, 4]), ([9, 4], 11, [8, 4])

11. iterace

Open:

([4, 3], 16, [5, 4]), ([6, 3], 15, [5, 4]), ([4, 4], 16, [5, 4]), ([7, 3], 15, [6, 4]), ([8, 3], 16, [7, 4]), ([9, 3], 17, [8, 4]), ([9, 5], 17, [8, 4]), ([3, 4], 14, [4, 5]), ([3, 6], 19, [4, 5]), ([4, 6], 18, [4, 5]), ([2, 5], 16, [3, 5]), ([2, 6], 22, [3, 5])

Closed:

([6, 6], 0, null), ([6, 5], 4, [6, 6]), ([7, 5], 5, [6, 6]), ([7, 6], 6, [6, 6]), ([5, 4], 7, [6, 5]), ([6, 4], 7, [6, 5]), ([7, 4], 7, [6, 5]), ([8, 4], 8, [7, 5]), ([4, 5], 10, [5, 4]), ([9, 4], 11, [8, 4]), ([3, 5], 13, [4, 5])

12. iterace

Open:

([4, 3], 16, [5, 4]), ([6, 3], 15, [5, 4]), ([4, 4], 16, [5, 4]), ([7, 3], 15, [6, 4]), ([8, 3], 16, [7, 4]), ([9, 3], 17, [8, 4]), ([9, 5], 17, [8, 4]), ([3, 6], 19, [4, 5]), ([4, 6], 18, [4, 5]), ([2, 5], 16, [3, 5]), ([2, 6], 22, [3, 5]), ([3, 3], 16, [3, 4])

Closed:

([6, 6], 0, null), ([6, 5], 4, [6, 6]), ([7, 5], 5, [6, 6]), ([7, 6], 6, [6, 6]), ([5, 4], 7, [6, 5]), ([6, 4], 7, [6, 5]), ([7, 4], 7, [6, 5]), ([8, 4], 8, [7, 5]), ([4, 5], 10, [5, 4]), ([9, 4], 11, [8, 4]), ([3, 5], 13, [4, 5]), ([3, 4], 14, [4, 5])

13. iterace

Open:

([4, 3], 16, [5, 4]), ([4, 4], 16, [5, 4]), ([7, 3], 15, [6, 4]), ([8, 3], 16, [7, 4]), ([9, 3], 17, [8, 4]), ([9, 5], 17, [8, 4]), ([3, 6], 19, [4, 5]), ([4, 6], 18, [4, 5]), ([2, 5], 16, [3, 5]), ([2, 6], 22, [3, 5]), ([3, 3], 16, [3, 4]), ([6, 2], 22, [6, 3]), ([7, 2], 20, [6, 3])

Closed:

([6, 6], 0, null), ([6, 5], 4, [6, 6]), ([7, 5], 5, [6, 6]), ([7, 6], 6, [6, 6]), ([5, 4], 7, [6, 5]), ([6, 4], 7, [6, 5]), ([7, 4], 7, [6, 5]), ([8, 4], 8, [7, 5]), ([4, 5], 10, [5, 4]), ([9, 4], 11, [8, 4]), ([3, 5], 13, [4, 5]), ([3, 4], 14, [4, 5]), ([6, 3], 15, [5, 4])

14. iterace

Open:

([4, 3], 16, [5, 4]), ([4, 4], 16, [5, 4]), ([8, 3], 16, [7, 4]), ([9, 3], 17, [8, 4]), ([9, 5], 17, [8, 4]), ([3, 6], 19, [4, 5]), ([4, 6], 18, [4, 5]), ([2, 5], 16, [3, 5]), ([2, 6], 22, [3, 5]), ([3, 3], 16, [3, 4]), ([6, 2], 22, [6, 3]), ([7, 2], 20, [6, 3]), ([8, 2], 21, [7, 3])

Closed:

([6, 6], 0, null), ([6, 5], 4, [6, 6]), ([7, 5], 5, [6, 6]), ([7, 6], 6, [6, 6]), ([5, 4], 7, [6, 5]), ([6, 4], 7, [6, 5]), ([7, 4], 7, [6, 5]), ([8, 4], 8, [7, 5]), ([4, 5], 10, [5, 4]), ([9, 4], 11, [8, 4]), ([3, 5], 13, [4, 5]), ([3, 4], 14, [4, 5]), ([6, 3], 15, [5, 4]), ([7, 3], 15, [6, 4])

15. iterace

Open:

([4, 4], 16, [5, 4]), ([8, 3], 16, [7, 4]), ([9, 3], 17, [8, 4]), ([9, 5], 17, [8, 4]), ([3, 6], 19, [4, 5]), ([4, 6], 18, [4, 5]), ([2, 5], 16, [3, 5]), ([2, 6], 22, [3, 5]), ([3, 3], 16, [3, 4]), ([6, 2], 22, [6, 3]), ([7, 2], 20, [6, 3]), ([8, 2], 21, [7, 3])

Closed:

([6, 6], 0, null), ([6, 5], 4, [6, 6]), ([7, 5], 5, [6, 6]), ([7, 6], 6, [6, 6]), ([5, 4], 7, [6, 5]), ([6, 4], 7, [6, 5]), ([7, 4], 7, [6, 5]), ([8, 4], 8, [7, 5]), ([4, 5], 10, [5, 4]), ([9, 4], 11, [8, 4]), ([3, 5], 13, [4, 5]), ([3, 4], 14, [4, 5]), ([6, 3], 15, [5, 4]), ([7, 3], 15, [6, 4]), ([4, 3], 16, [5, 4])

16. iterace

Open:

([8, 3], 16, [7, 4]), ([9, 3], 17, [8, 4]), ([9, 5], 17, [8, 4]), ([3, 6], 19, [4, 5]), ([4, 6], 18, [4, 5]), ([2, 5], 16, [3, 5]), ([2, 6], 22, [3, 5]), ([3, 3], 16, [3, 4]), ([6, 2], 22, [6, 3]), ([7, 2], 20, [6, 3]), ([8, 2], 21, [7, 3])

Closed:

([6, 6], 0, null), ([6, 5], 4, [6, 6]), ([7, 5], 5, [6, 6]), ([7, 6], 6, [6, 6]), ([5, 4], 7, [6, 5]), ([6, 4], 7, [6, 5]), ([7, 4], 7, [6, 5]), ([8, 4], 8, [7, 5]), ([4, 5], 10, [5, 4]), ([9, 4], 11, [8, 4]), ([3, 5], 13, [4, 5]), ([3, 4], 14, [4, 5]), ([6, 3], 15, [5, 4]), ([7, 3], 15, [6, 4]), ([4, 3], 16, [5, 4]), ([4, 4], 16, [5, 4])

17. iterace

Open:

([9, 3], 17, [8, 4]), ([9, 5], 17, [8, 4]), ([3, 6], 19, [4, 5]), ([4, 6], 18, [4, 5]), ([2, 5], 16, [3, 5]), ([2, 6], 22, [3, 5]), ([3, 3], 16, [3, 4]), ([6, 2], 22, [6, 3]), ([7, 2], 20, [6, 3]), ([8, 2], 21, [7, 3]), ([9, 2], 22, [8, 3])

Closed:

([6, 6], 0, null), ([6, 5], 4, [6, 6]), ([7, 5], 5, [6, 6]), ([7, 6], 6, [6, 6]), ([5, 4], 7, [6, 5]), ([6, 4], 7, [6, 5]), ([7, 4], 7, [6, 5]), ([8, 4], 8, [7, 5]), ([4, 5], 10, [5, 4]), ([9, 4], 11, [8, 4]), ([3, 5], 13, [4, 5]), ([3, 4], 14, [4, 5]), ([6, 3], 15, [5, 4]), ([7, 3], 15, [6, 4]), ([4, 3], 16, [5, 4]), ([4, 4], 16, [5, 4]), ([8, 3], 16, [7, 4])

18. iterace

Open:

([9, 3], 17, [8, 4]), ([9, 5], 17, [8, 4]), ([3, 6], 19, [4, 5]), ([4, 6], 18, [4, 5]), ([2, 6], 22, [3, 5]), ([3, 3], 16, [3, 4]), ([6, 2], 22, [6, 3]), ([7, 2], 20, [6, 3]), ([8, 2], 21, [7, 3]), ([9, 2], 22, [8, 3]), ([1, 4], 25, [2, 5]), ([1, 5], 19, [2, 5]), ([1, 6], 25, [2, 5])

Closed:

([6, 6], 0, null), ([6, 5], 4, [6, 6]), ([7, 5], 5, [6, 6]), ([7, 6], 6, [6, 6]), ([5, 4], 7, [6, 5]), ([6, 4], 7, [6, 5]), ([7, 4], 7, [6, 5]), ([8, 4], 8, [7, 5]), ([4, 5], 10, [5, 4]), ([9, 4], 11, [8, 4]), ([3, 5], 13, [4, 5]), ([3, 4], 14, [4, 5]), ([6, 3], 15, [5, 4]), ([7, 3], 15, [6, 4]), ([4, 3], 16, [5, 4]), ([4, 4], 16, [5, 4]), ([8, 3], 16, [7, 4]), ([2, 5], 16, [3, 5])

19. iterace

Open:

Closed:

20. iterace

Open:

Closed:

21. iterace

Open:

Closed: